

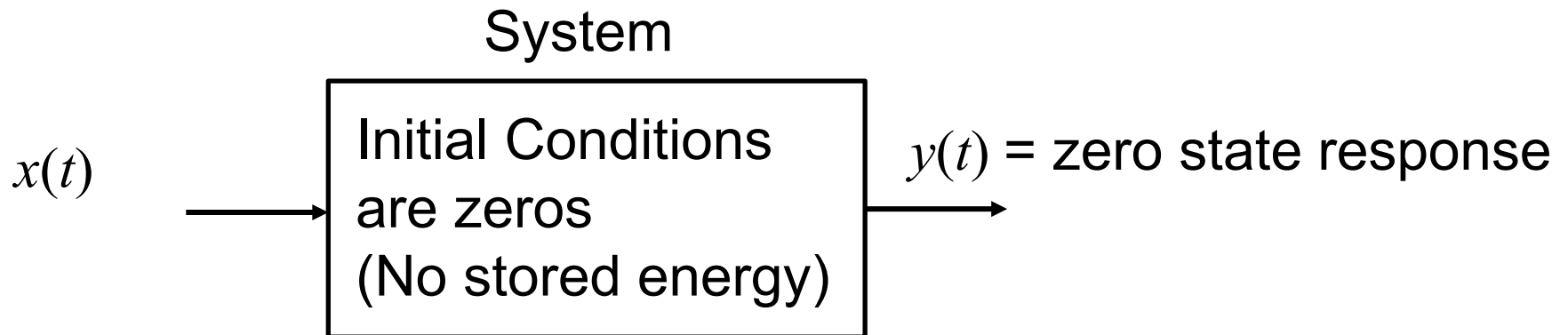
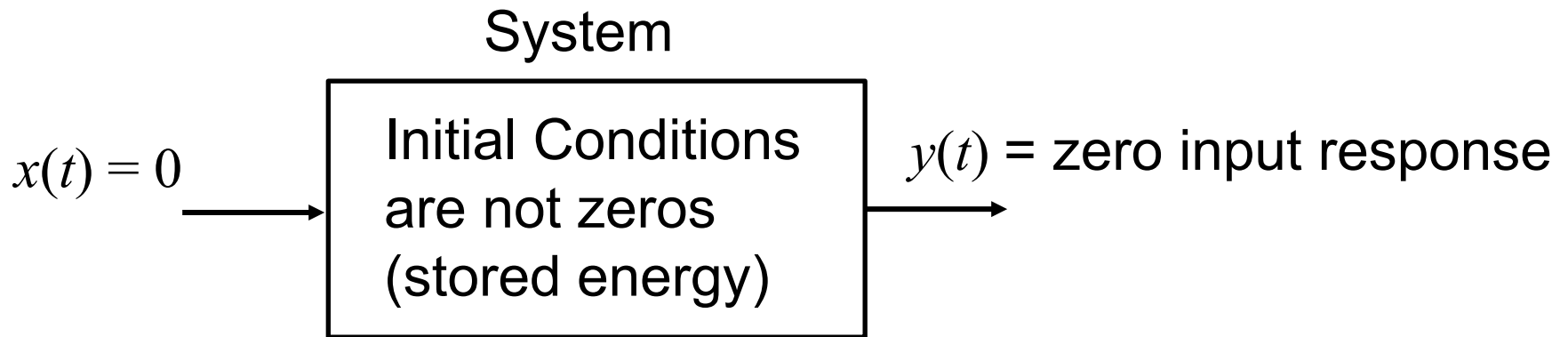
Time Domain Analysis of Linear Systems Ch2

University of Central Oklahoma
Dr. Mohamed Bingabr

Outline

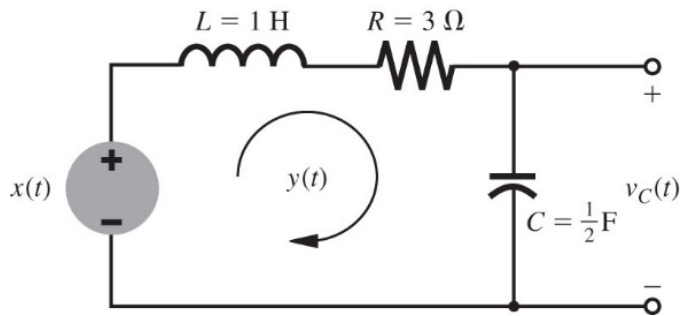
- Zero-input Response
- Impulse Response $h(t)$
- Convolution
- Zero-State Response
- System Stability

System Response



Total response = Zero input response + Zero state response

System Response

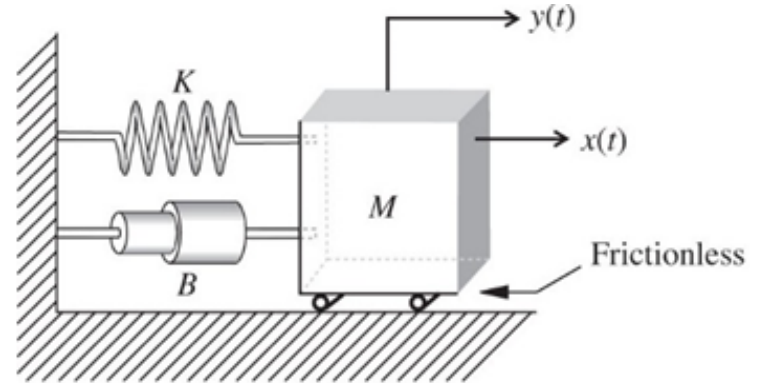


$$\frac{d^2 y}{dt^2} + \frac{R}{L} \frac{dy}{dt} + \frac{1}{LC} y = \frac{1}{L} \frac{dx}{dt}$$

$$\left(D^2 + \frac{R}{L} D + \frac{1}{LC} \right) y = \frac{1}{L} D x$$

For N order system

$$(D^N + a_1 D^{N-1} + a_2 D^{N-2} + \dots + a_N) y(t) = (b_0 D^M + b_1 D^{M-1} + b_2 D^{M-2} + \dots + b_M) x(t)$$



$$\frac{d^2 y}{dt^2} + \frac{B}{M} \frac{dy}{dt} + \frac{k}{M} y = \frac{1}{M} x$$

$$\left(D^2 + \frac{B}{M} D + \frac{k}{M} \right) y = \frac{1}{M} x$$

What is the zero-input response?

Zero-Input Response

$$(D^N + a_1 D^{N-1} + a_2 D^{N-2} + \cdots + a_N)y(t) = 0$$

A solution to the above differential equation is $y(t) = ce^{\lambda t}$

$$Dy = \lambda ce^{\lambda t} \qquad D^2 y = \lambda^2 ce^{\lambda t} \qquad D^N y = \lambda^N ce^{\lambda t}$$

$$\underbrace{(\lambda^N + a_1 \lambda^{N-1} + a_2 \lambda^{N-2} + \cdots + a_N)}_{\text{Characteristic Polynomial } Q(\lambda)} ce^{\lambda t} = 0$$

Characteristic Polynomial $Q(\lambda)$

Characteristic Equation $Q(\lambda) = 0$

$$\lambda^N + a_1 \lambda^{N-1} + a_2 \lambda^{N-2} + \cdots + a_N = 0$$

Zero-Input Response

Next step is to solve the characteristic equation

$$\lambda^N + a_1\lambda^{N-1} + a_2\lambda^{N-2} + \dots + a_N = 0$$

$$(\lambda - \lambda_1)(\lambda - \lambda_2) \dots (\lambda - \lambda_N) = 0$$

If the **characteristic roots** λ s are distinct real, then possible solutions to the differential equation are

$$c_1 e^{\lambda_1 t} \quad c_2 e^{\lambda_2 t} \quad c_N e^{\lambda_N t}$$

A general solution of the zero-input response is

$$y(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t} + \dots + c_N e^{\lambda_N t}$$

Zero-Input Response

The **zero-input response** is

$$y(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t} + \dots + c_N e^{\lambda_N t}$$

Characteristic Roots: $\lambda_1 \dots \lambda_N$

- Also called the natural frequencies or eigenvalues
- For stable system all $\lambda \leq 0$
- Can be real distinct, repeated roots, and/or complex roots

Characteristic Modes: $c_1 e^{\lambda_1 t} \dots c_N e^{\lambda_N t}$, determine the system's behavior

The constants $c_1, \dots, c_n$ are arbitrary constants that will be determined by the initial conditions (initial states of the system)

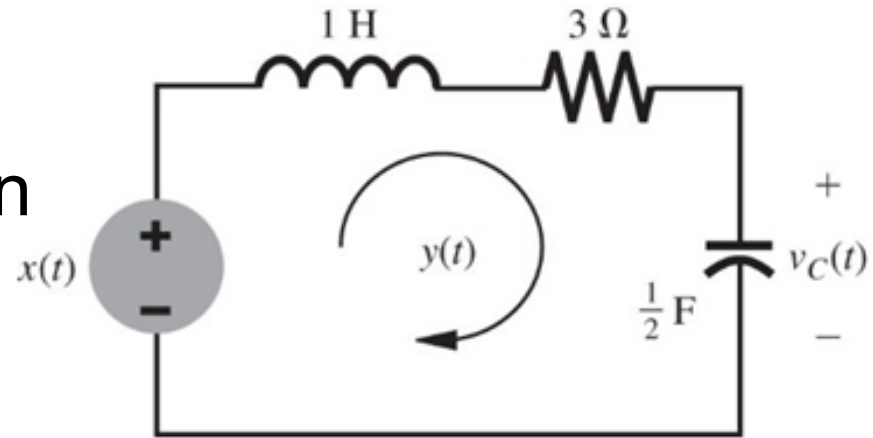
Example

For the LTI system described by the following differential equation $(D^2 + 3D + 2) y(t) = D x(t)$

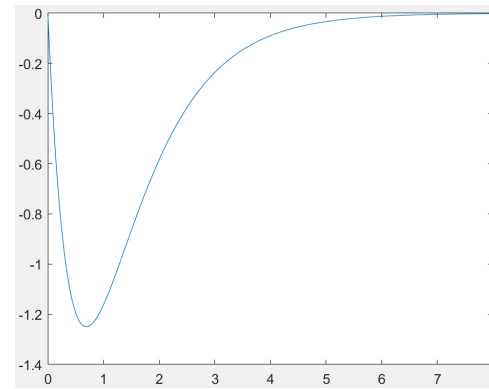
The initial conditions are $y(0) = 0$ and $\dot{y}(0) = -5$

Find

- a) The characteristic equation
- b) The characteristic roots
- c) The characteristic modes
- d) The zero-input response



Answer: $y(t) = -5e^{-t} + 5e^{-2t}$



Repeated Characteristic Roots

If the characteristic equation

$$(\lambda - \lambda_1)(\lambda - \lambda_2) \dots (\lambda - \lambda_N) = 0$$

has this form

$$(\lambda - \lambda_1)^N = 0,$$

then the N characteristic roots has the same value λ_1 .

The zero-input solution will be

$$y(t) = c_1 e^{\lambda_1 t} + c_2 t e^{\lambda_1 t} + c_3 t^2 e^{\lambda_1 t} + \dots + c_N t^{N-1} e^{\lambda_1 t}$$

Example

For the LTI system described by the following differential equation $(D^2 + 6D + 9) y(t) = (3D + 5) x(t)$
The initial conditions are $y(0) = 3$ and $\dot{y}(0) = -7$

Find

- a) The characteristic equation
- b) The characteristic roots
- c) The characteristic modes
- d) The zero-input response

Answer: $y(t) = 3e^{-3t} + 2te^{-3t}$

Complex Characteristic Roots

If the characteristic equation

$$(\lambda - \lambda_1)(\lambda - \lambda_2) \dots (\lambda - \lambda_N) = 0$$

has a complex characteristic root $\sigma + j\omega$ then its conjugate $\sigma - j\omega$ is also a characteristic root. This is necessary for the system to be physically realizable.

The zero-input solution for a pair of conjugate roots is

$$y(t) = c_1 e^{(\sigma + j\omega)t} + c_2 e^{(\sigma - j\omega)t}$$

If c_1 and c_2 are complex, then c_2 is conjugate of c_1 .

$$y(t) = 0.5c e^{j\theta} e^{(\sigma + j\omega)t} + 0.5c e^{-j\theta} e^{(\sigma - j\omega)t}$$

$$y(t) = c e^{\sigma t} \cos(\omega t + \theta)$$

Example

For the LTI system described by the following differential equation $(D^2 + 4D + 40) y(t) = (D + 2) x(t)$
The initial conditions are $y(0) = 2$ and $\dot{y}(0) = 16.78$

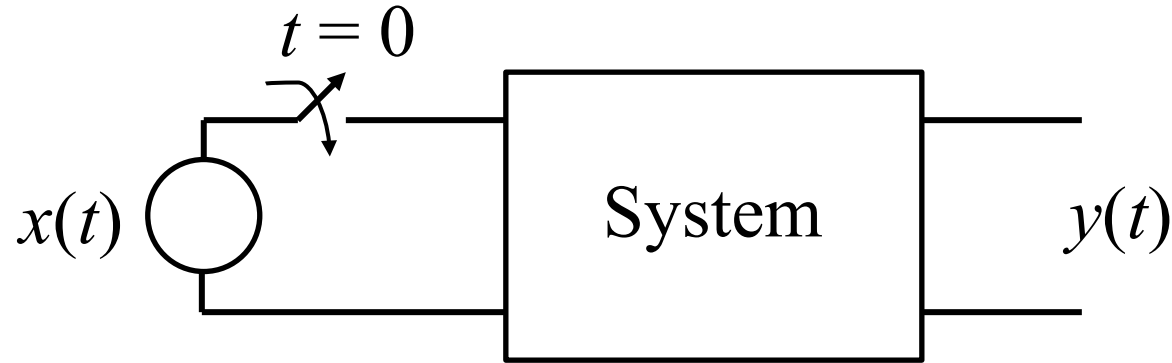
Find

- a) The characteristic equation
- b) The characteristic roots
- c) The characteristic modes
- d) The zero-input response

Answer: $y(t) = 4e^{-2t} \cos\left(6t - \frac{\pi}{3}\right)$ or

$$y(t) = -4e^{-2t} \cos\left(6t + \frac{2\pi}{3}\right)$$

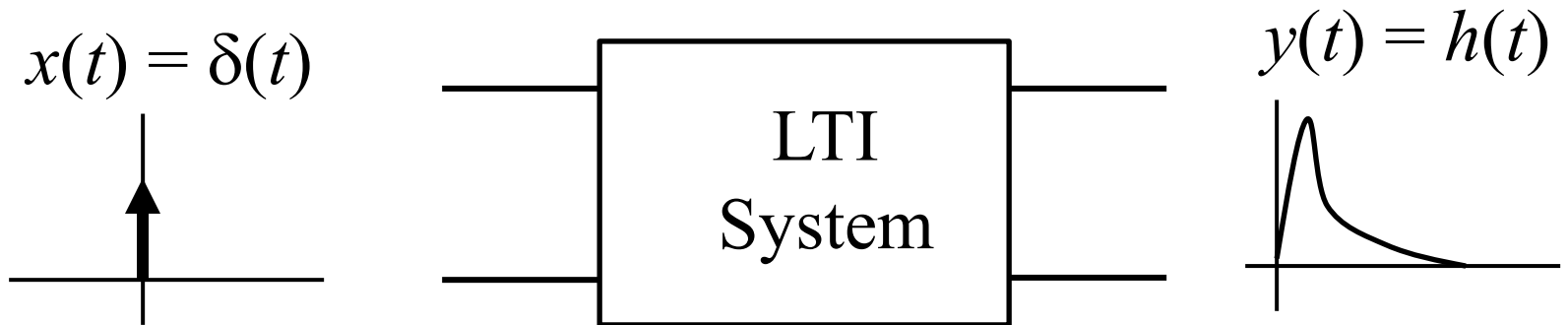
The Meaning of $t = 0^-$ and $t = 0^+$



$y(0^-)$: At $t = 0^-$ the input is not applied to the system yet so the output is due only to the initial conditions.

$y(0^+)$: At $t = 0^+$ the input is applied to the system, so the output is due the input and the initial conditions.

Impulse Response $h(t)$

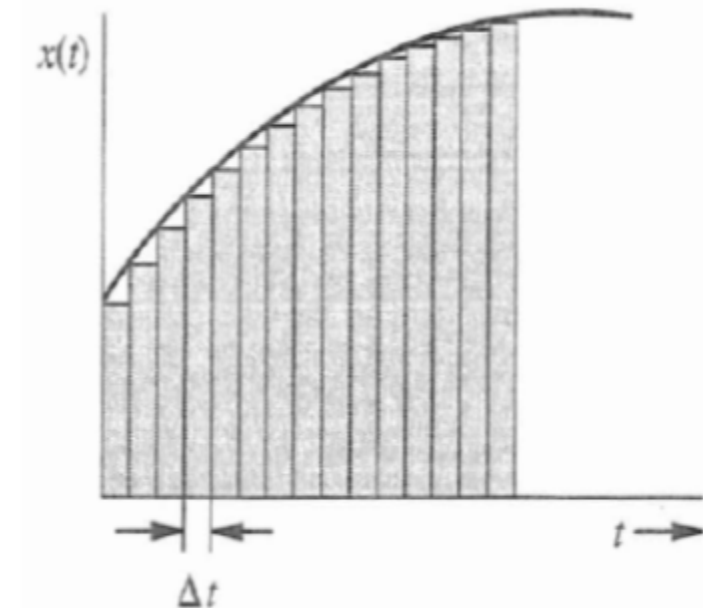


Unit Impulse Response $h(t)$

Reveal system behavior

Depends on the system internal characteristic modes

Helps in finding system response to any input $x(t)$

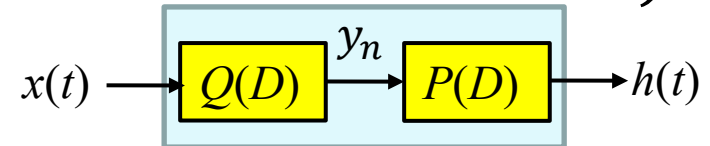


Derivation of Impulse Response $h(t)$

1- Find the differential equation of the system

$$(D^N + a_1 D^{N-1} + \dots + a_N)y = (b_0 D^M + b_1 D^{M-1} + \dots + b_M)x$$

$$Q(D)y(t) = P(D)x(t)$$



2- Find the natural response $y_n(t)$ using the same steps used to find the zero-input response.

$$y_n(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t} + \dots + c_N e^{\lambda_N t}$$

3- To find the constants c , set all initial conditions to zeros except $N-1$ derivative, set it to equal 1.

$$y_n(0) = \dot{y}_n(0) = \ddot{y}_n(0) = \dots = y_n^{N-2}(0) = 0 \quad y_n^{N-1}(0) = 1$$

4- $h(t) = [P(D)y_n(t)]u(t)$

If $M = N$ then $h(t) = b_0 \delta(t) + [P(D)y_n(t)]u(t)$

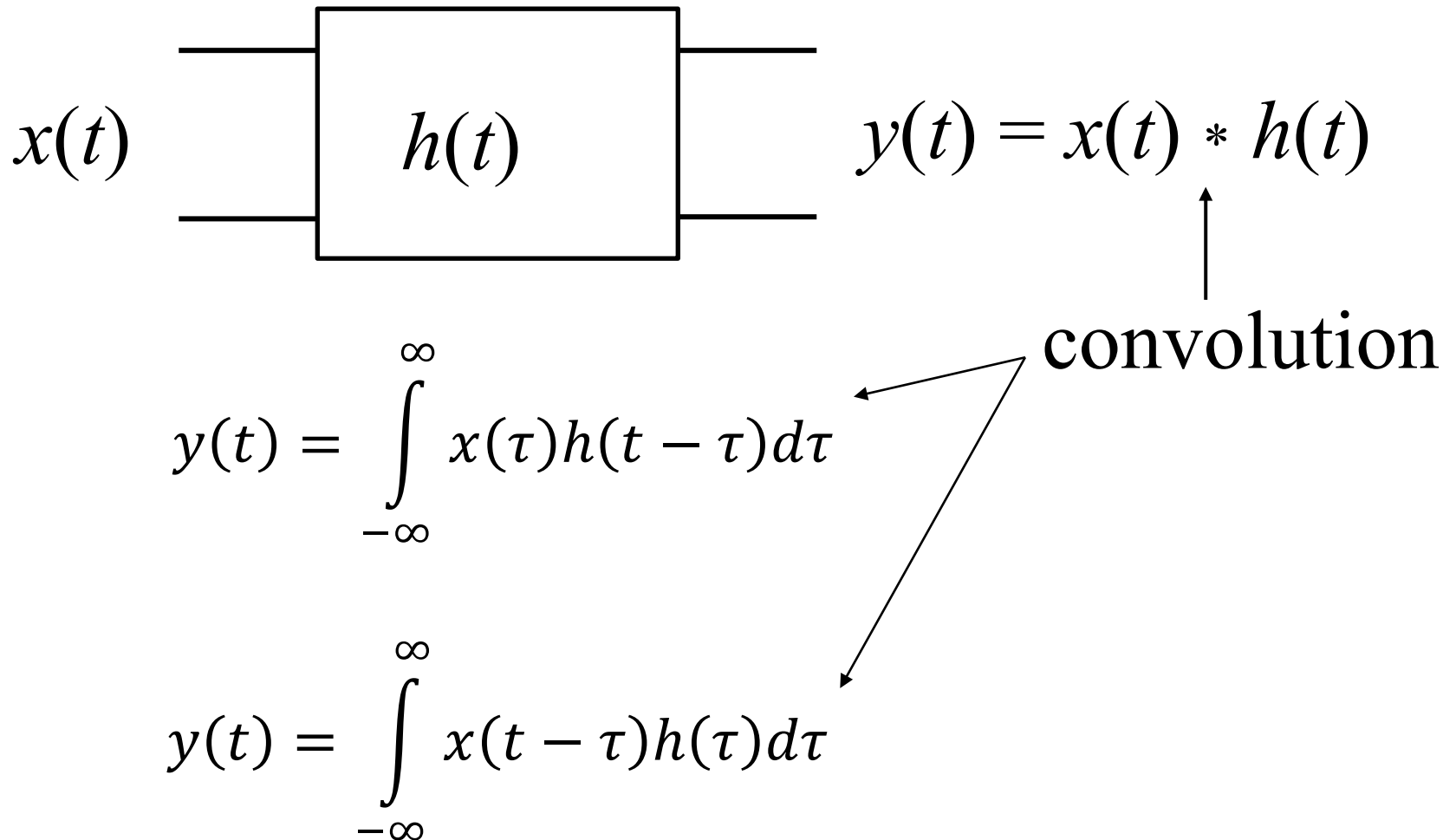
Example

Determine the impulse response $h(t)$ for the system

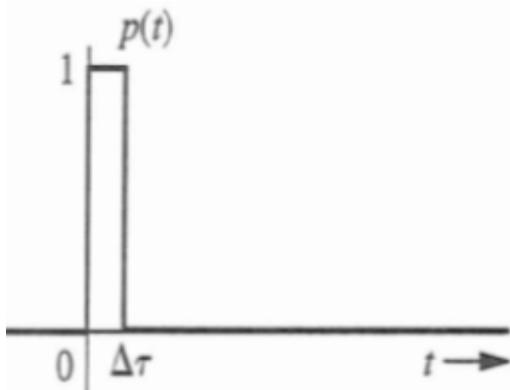
$$(D^2 + 3D + 2) y(t) = (D + 3) x(t)$$

Answer: $y(t) = (2e^{-t} - e^{-2t})u(t)$

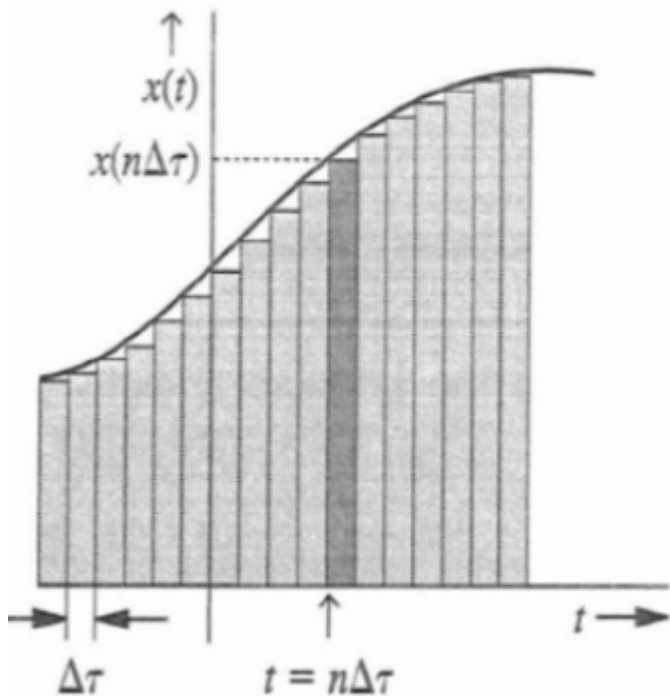
Zero-State Response



Zero-State Response



Any signal can be represented as a train of pulses of different amplitudes and at different locations



$$x(t) = \lim_{\Delta\tau \rightarrow 0} \sum_n x(n\Delta\tau) p(t - n\Delta\tau)$$

$$x(t) = \lim_{\Delta\tau \rightarrow 0} \sum_n x(n\Delta\tau) \frac{p(t - n\Delta\tau)}{\Delta\tau} \Delta\tau$$

$$x(t) = \lim_{\Delta\tau \rightarrow 0} \sum_n x(n\Delta\tau) \delta(t - n\Delta\tau) \Delta\tau$$

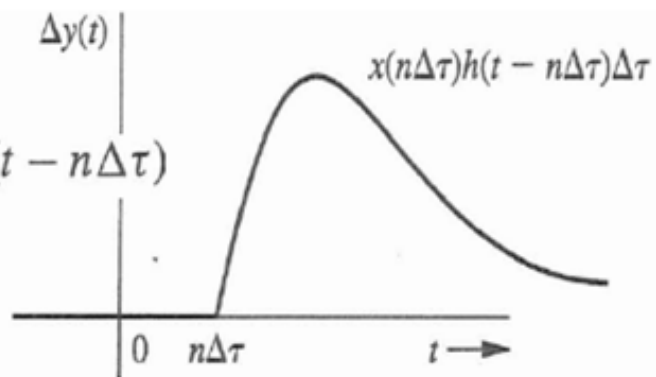
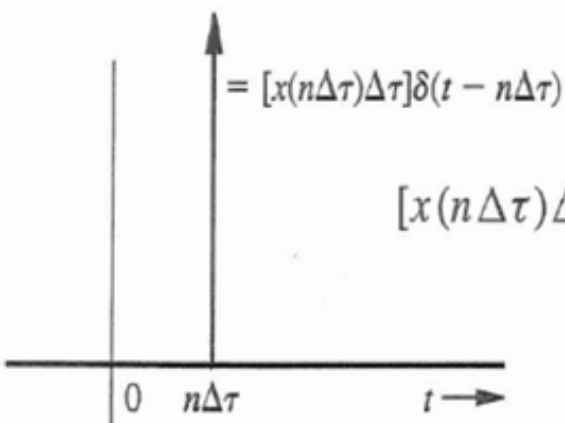
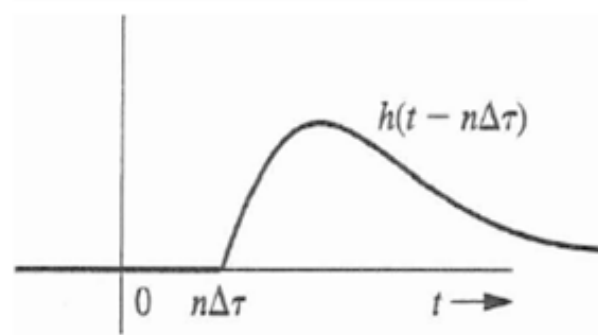
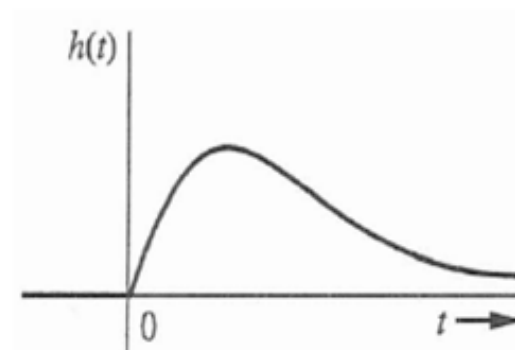
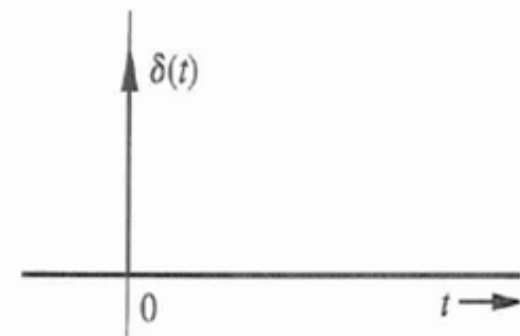
Zero-State Response

input $\Rightarrow$ output

$\delta(t) \Rightarrow h(t)$

$\delta(t - n\Delta\tau) \Rightarrow h(t - n\Delta\tau)$

$[x(n\Delta\tau)\Delta\tau]\delta(t - n\Delta\tau) \Rightarrow [x(n\Delta\tau)\Delta\tau]h(t - n\Delta\tau)$



Zero-State Response

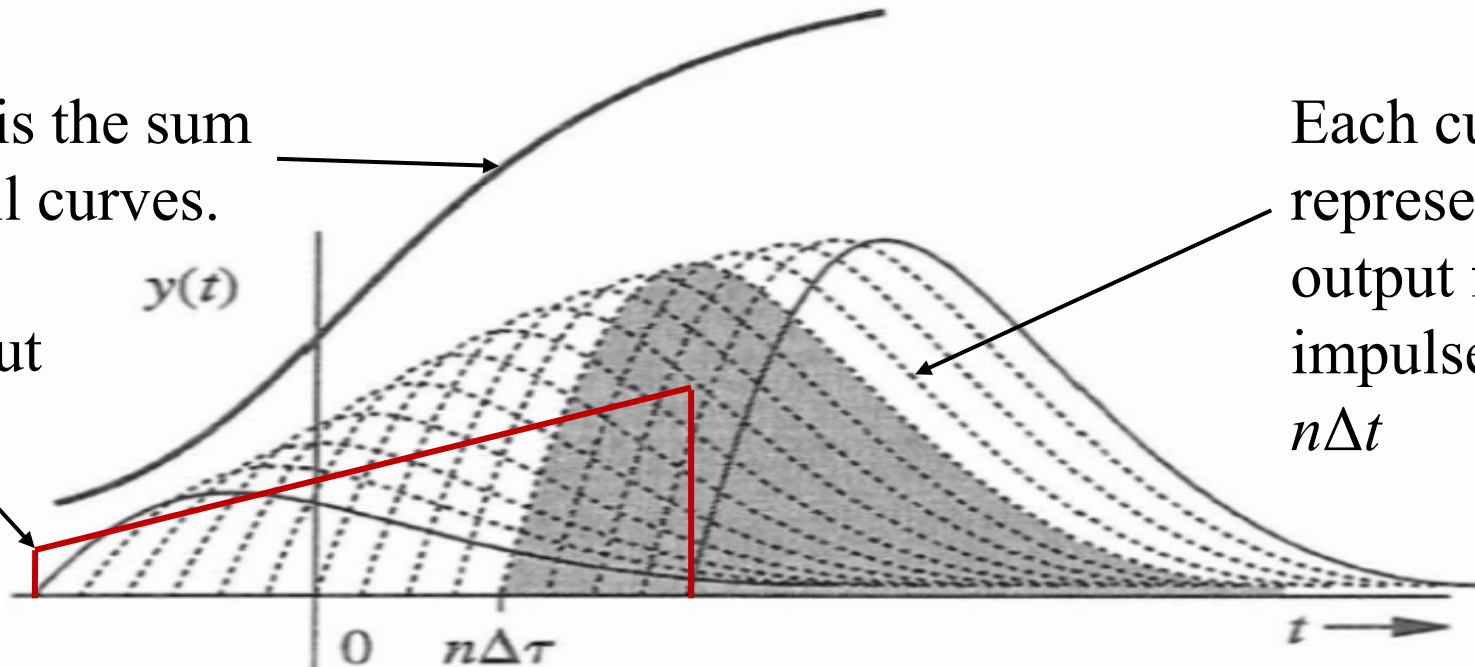
$$\lim_{\Delta\tau \rightarrow 0} \sum_n x(n\Delta\tau) \delta(t - n\Delta\tau) \Delta\tau \quad \Rightarrow \quad \lim_{\Delta\tau \rightarrow 0} \sum_n x(n\Delta\tau) h(t - n\Delta\tau) \Delta\tau$$

$x(t)$ $y(t)$

$y(t)$ is the sum of all curves.

Each curve represents the output for impulse input at $n\Delta t$

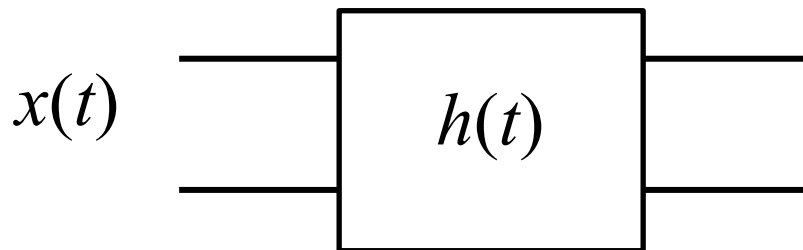
Input $x(t)$



$$y(t) = \lim_{\Delta\tau \rightarrow 0} \sum_n x(n\Delta\tau) h(t - n\Delta\tau) \Delta\tau = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$$

Zero-State Response

The Convolution Integral



$$y(t) = x(t) * h(t)$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau$$

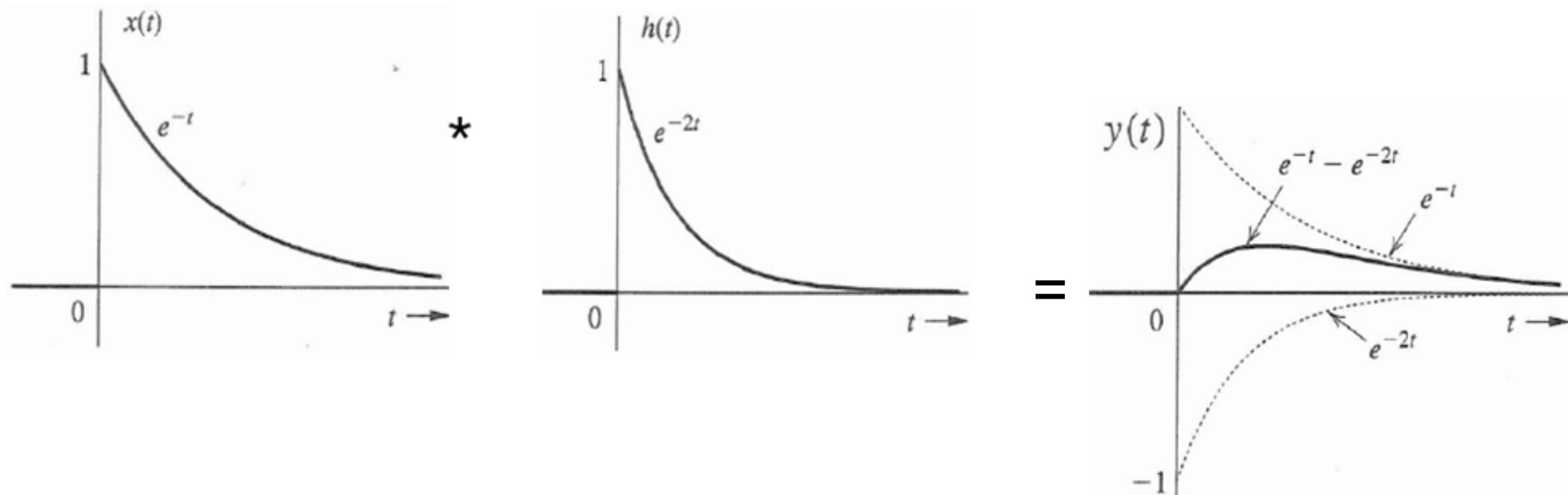
$$y(t) = h(t) * x(t)$$

$$y(t) = \int_{-\infty}^{\infty} h(\tau)x(t - \tau)d\tau$$

Example

For a LTI system with impulse response $h(t) = e^{-2t} u(t)$, determine the response $y(t)$ for the input $x(t) = e^{-t} u(t)$.

Answer: $y(t) = (e^{-t} - e^{-2t})u(t)$



The Convolution Properties

Commutative: $x_1 * x_2 = x_2 * x_1$

Associative: $x_1 * [x_2 * x_3] = [x_1 * x_2] * x_3$

Distributive: $x_1 * [x_2 + x_3] = [x_1 * x_2] + [x_1 * x_3]$

Impulse Convolution: $x(t) * \delta(t) = \int x(\tau) \delta(t - \tau) d\tau = x(t)$

The Convolution Properties

Shift Property:

if $x_1(t) * x_2(t) = c(t)$

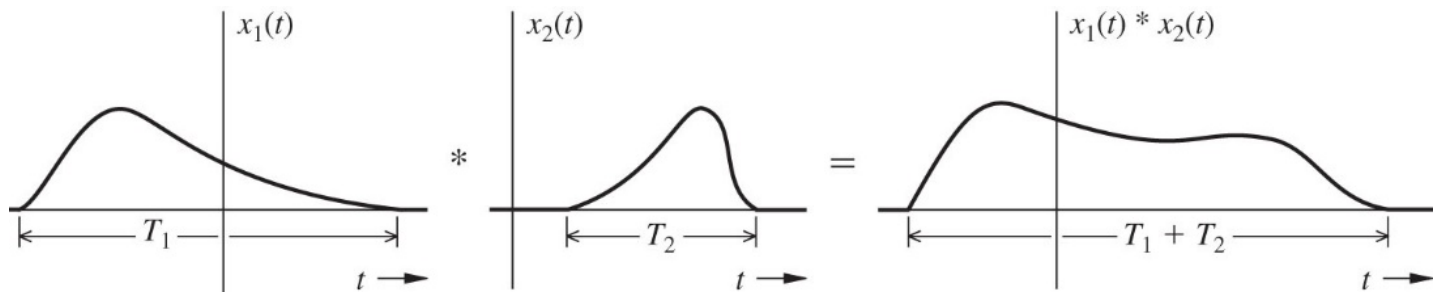
then $x_1(t - T) * x_2(t) = x_1(t) * x_2(t - T) = c(t - T)$

also $x_1(t - T_1) * x_2(t - T_2) = c(t - T_1 - T_2)$

Width Property:

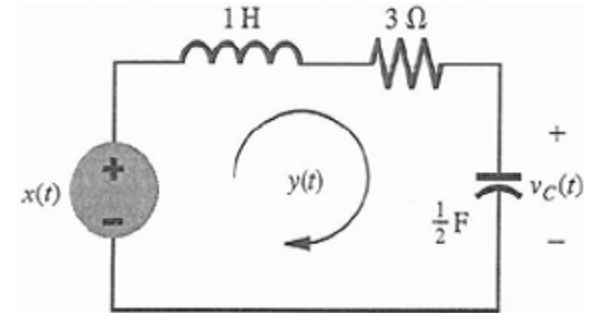
If the width of $x_1(t)$ is T_1 and the width of $x_2(t)$ is T_2

then the width of $x_1(t) * x_2(t)$ is $T_1 + T_2$



Example

Find the total response for the system $D^2y + 3Dy + 2y = Dx$ for input $x(t)=10e^{-3t}u(t)$ with initial condition $y(0) = 0$ and $\dot{y}(0) = -5$



Answer

$$y(t) = \underbrace{\left(-5e^{-t} + 5e^{-2t}\right)}_{\text{Zero-input Response}} + \underbrace{\left(-5e^{-t} + 20e^{-2t} - 15e^{-3t}\right)}_{\text{Zero-state Response}} \quad \text{For } t \geq 0$$

$$y(t) = \underbrace{\left(-10e^{-t} + 25e^{-2t}\right)}_{\text{Natural Response}} + \underbrace{\left(-15e^{-3t}\right)}_{\text{Forced Response}}$$

System Stability

- External Stability (BIBO)

- If the input is bounded, then the output is bounded

- $\int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty$

- Internal Stability (Asymptotic)

- If and only if all the characteristic roots are in the LHP

- Unstable if, and only if, one or both of the following conditions exist:

- At least one root is in the RHP
 - There are repeated roots on the imaginary axis

- Marginally stable if, and only if, there are no roots in the RHP, and there are some unrepeated roots on the imaginary axis.

Example

Investigate the asymptotic & BIBO stability of the following systems:

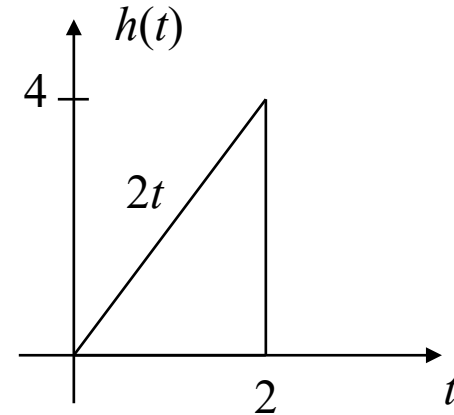
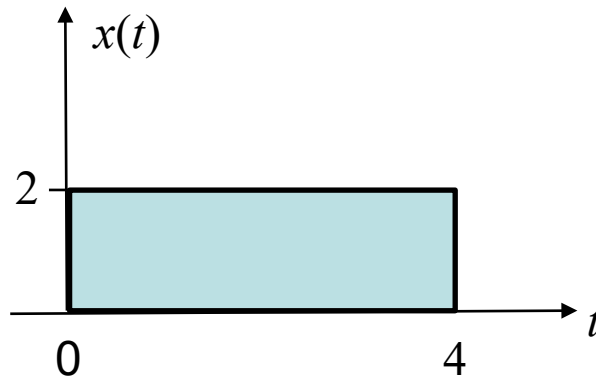
a) $(D+1)(D^2 + 4D + 8) y(t) = (D - 3) x(t)$

b) $(D-1)(D^2 + 4D + 8) y(t) = (D + 2) x(t)$

c) $(D+2)(D^2 + 4) y(t) = (D^2 + D + 1) x(t)$

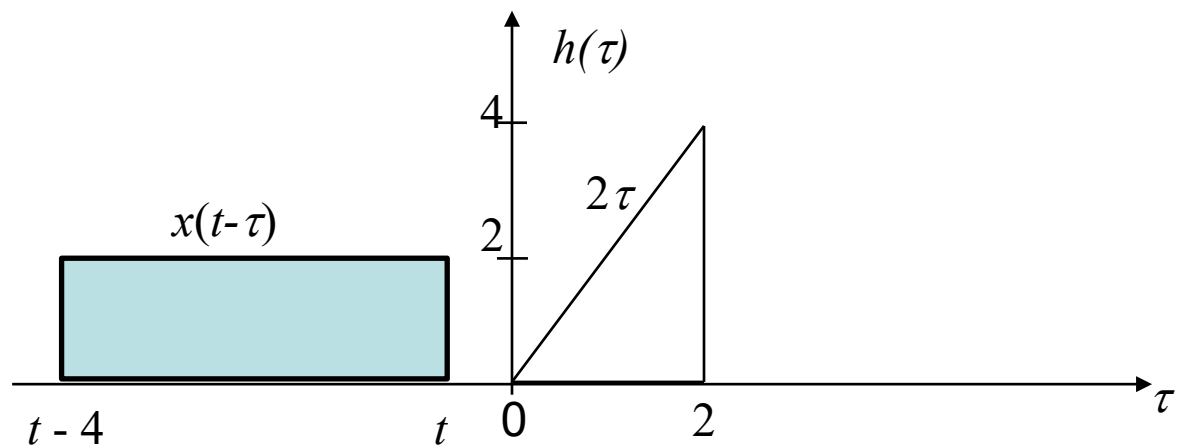
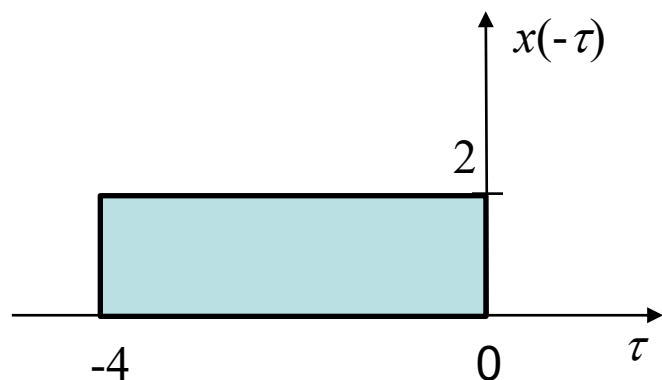
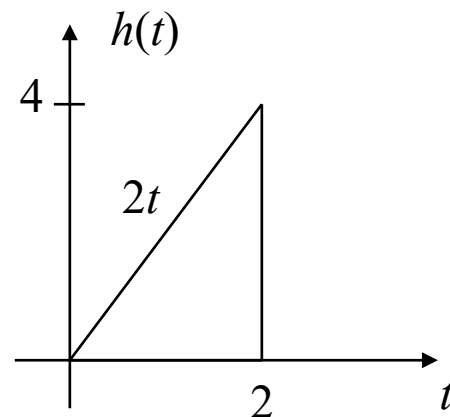
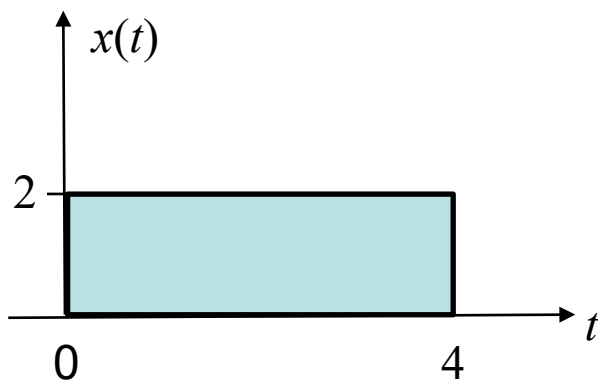
Convolution Example

Find the zero-state response $y(t)$ of the system described by the impulse response $h(t)$ for an input $x(t)$, shown below.



Convolution Example

Find the zero-state response $y(t)$ of the system described by the impulse response $h(t)$ for an input $x(t)$, shown below.

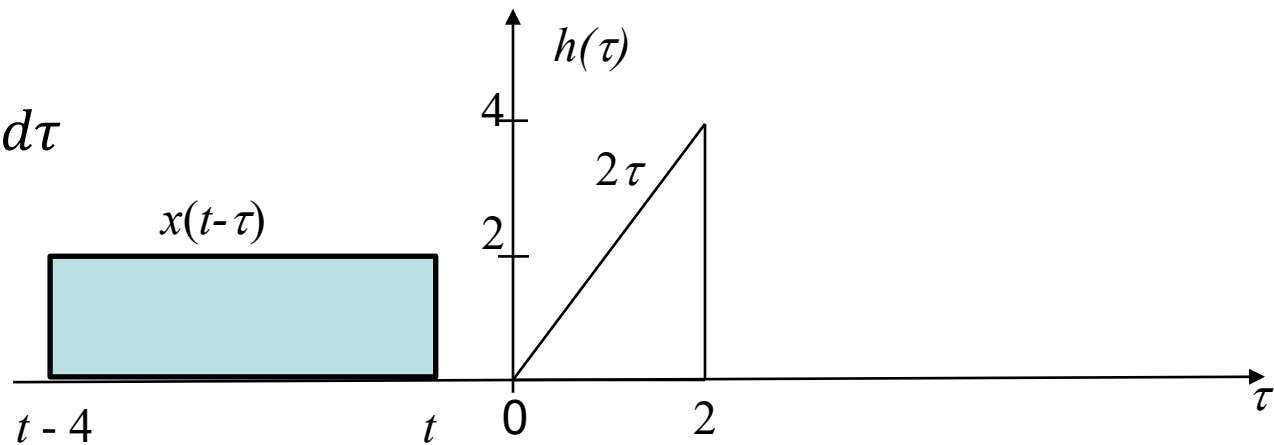


$$y(t) = \int_{-\infty}^{\infty} x(t - \tau)h(\tau)d\tau$$

Convolution Demonstration

$$y(t) = \int_{-\infty}^{\infty} x(t - \tau)h(\tau)d\tau$$

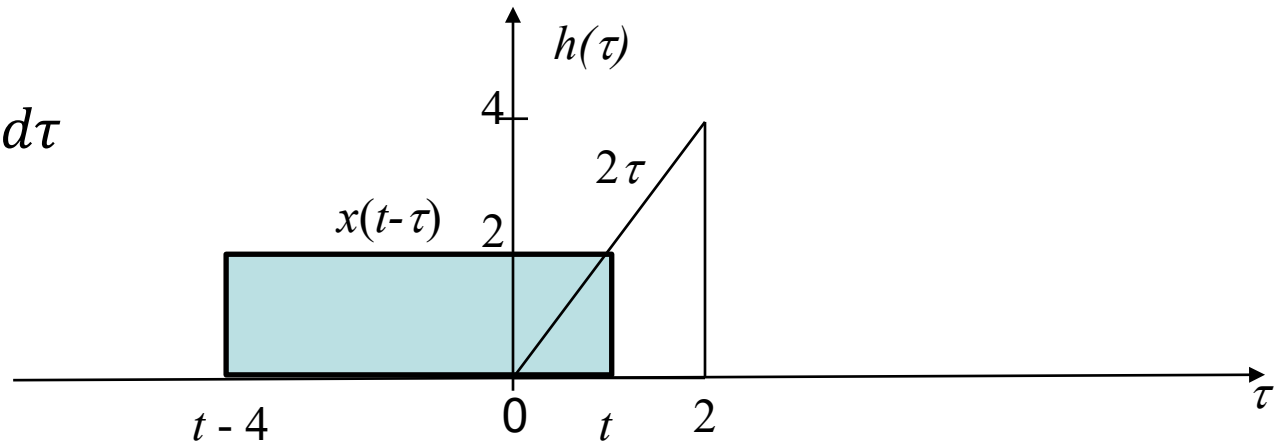
$y(t)$ for $t < 0$



Convolution Demonstration

$$y(t) = \int_{-\infty}^{\infty} x(t - \tau)h(\tau)d\tau$$

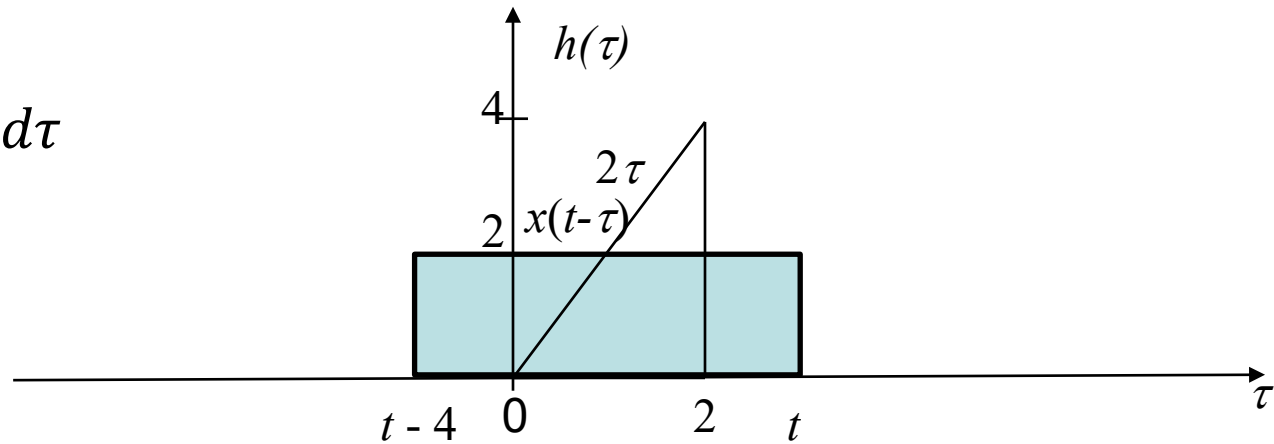
$y(t)$ for $0 < t < 2$



Convolution Demonstration

$$y(t) = \int_{-\infty}^{\infty} x(t - \tau)h(\tau)d\tau$$

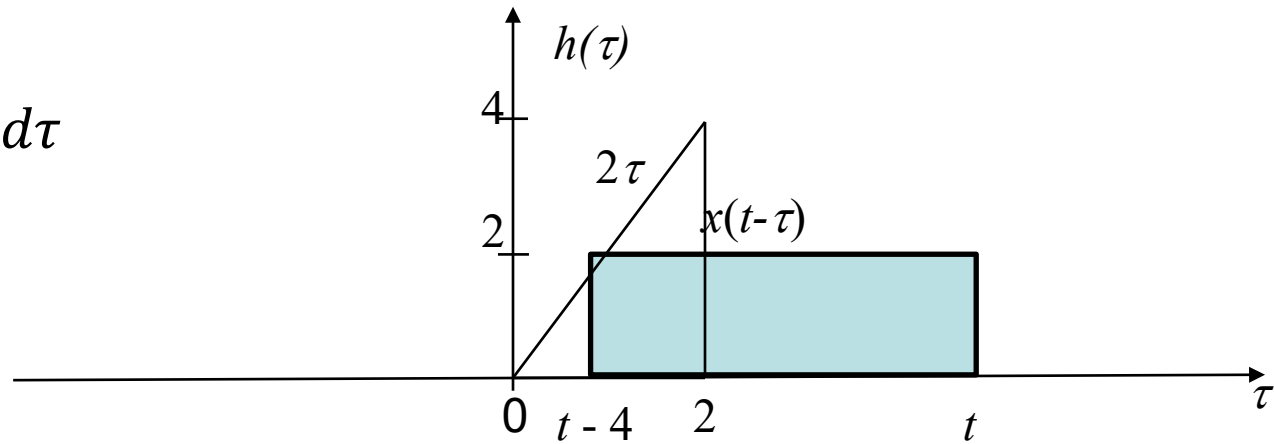
$y(t)$ for $2 < t < 4$



Convolution Demonstration

$$y(t) = \int_{-\infty}^{\infty} x(t - \tau)h(\tau)d\tau$$

$y(t)$ for $4 < t < 6$



Convolution Demonstration

$$y(t) = \int_{-\infty}^{\infty} x(t - \tau)h(\tau)d\tau$$

$y(t)$ for $t > 6$

